# EventHub

## Integration and System Testing Report

Part 5 · Call 5 design review · Selenium · Python / Flask / SQLite

John Bachta, with Claude (Sonnet 5.5, in Claude Code) as the GenAI tool | November 1, 2026

This report builds on the Part 4 package (`EventHub_Team5_Part4`). It covers what Call 5 changed in our requirements, model and code, how the GenAI-written Selenium tests behaved when they were first run, and what the final test run shows. Where something is unverified or still depends on the customer, it says so.

### At a glance

| Deliverable | What is included | Grade |
|---|---|---|
| 1. Design changes after Call 5 | Findings checked against our real build, 19 new and 3 superseded EARS requirements, new Request Rate Limiter component, purchase-token design, three ADRs, risk register | 20% |
| 2. How the test process went | First run as generated (110 passed, 27 failed), cause of every failure, what changed, what the tests found, mutation check, load measurement | 60% |
| 3. Tests that run on the final app | One command: `python run_all_tests.py`. 281 tests, all passing on the final code | 15% |
| 4. The test cases | `tests_system/` (137 Selenium and load cases), `tests/` (144), `docs/system-test-catalog.md`, `docs/test-plan.md` | 5% |

**Verified status and limits**

**Final run (Nov 1, 2026, Linux sandbox, headless Chrome 153):** 144 unit and integration tests and 137 browser and load tests passed, 0 failed, 0 skipped. Ruff lint and format, the course EARS/C4 gate (107 requirements) and the requirement-to-test check passed. 20 of 20 deliberate bugs were caught.

**Not verified:** the GitHub Actions workflow (re-created, never run on GitHub); macOS, Windows, Firefox, Safari or real phones; the Structurizr CLI (not installed, so the DSL was checked only by the course script).

**Still open:** Kenji's two findings are not formally accepted by anyone (they need a named signature). Marcus has not supplied the performance sentence. The discount percentage, email-change policy and rate-limit numbers are unknown. Part 5 did not decide them on the customer's behalf.

## 1. What changed after Call 5 (20%)

Before changing anything I ran the Call 5 scenarios against the Part 4 build exactly as delivered, because the call describes a build we do not fully share. The Part 4 code still passes its own 59 tests. It does not pass its own quality gates as the Part 4 report describes them: `ruff format --check` wants to rewrite 20 files, and `identity.py` has no PBKDF2 pin although the report says it was added. Both are fixed in Part 5.

## 1.1 Each finding, checked against our build

| Call 5 finding | What we found when we checked | Verdict |
|---|---|---|
| Double buy (Thuy): two orders, one, or reject? Nothing defines it. | Double-clicking the checkout button or triple-clicking a card made one order (the browser appears not to send the second submit while the page is loading; I did not investigate why). The same form sent twice made two orders and took 4 tickets for one 2-ticket purchase. | Real bug at the server. Defined and fixed (ADR-006) |
| Discount survives an email change (Kenji) | We could not reproduce it. No screen, route or function changes an email. Eligibility is judged from the current email on every price. The real hole is different: nobody checks that a registrant owns an `@usf.edu` mailbox (Q3, open since Part 1). | Not reproducible. Property written down and tested; real hole logged as RISK-001 |
| Request speed (Kenji): 200 requests a minute look like an enthusiastic student | 200 of 200 requests accepted, 150 orders created, none refused. | Real. Fixed with provisional numbers (ADR-007) |
| Confirmation page is "sketchy" (Darnell's students) | The page showed "Currency / locked exchange rate: EUR / 0.92" and "Total", with no sentence saying what was charged and no USD price. See Figure 1. | Real. Reworded (purchase-029/030) |
| Default currency wrong and unnoticed (Darnell) | Thuy said it is "chosen at registration". It is not. Registration had no currency field; every account silently defaulted to USD, and `/currency` existed with no screen to reach it. Quick-buy used it without showing it. | Call 5 description did not match our build. Fixed |
| Coverage is not behaviour (Thuy, Priya) | Part 4 tests carried no requirement IDs. Nine active requirements had no test at all (for example "no email service" and "rates come from the database"). | Real. Tag rule, traceability check, mutation check |
| No performance number (Thuy) | `system-006` is still `[TBD: Q13]`. We cannot decide the customer's number. | Not ours to fix. Measured instead; RISK-003 |

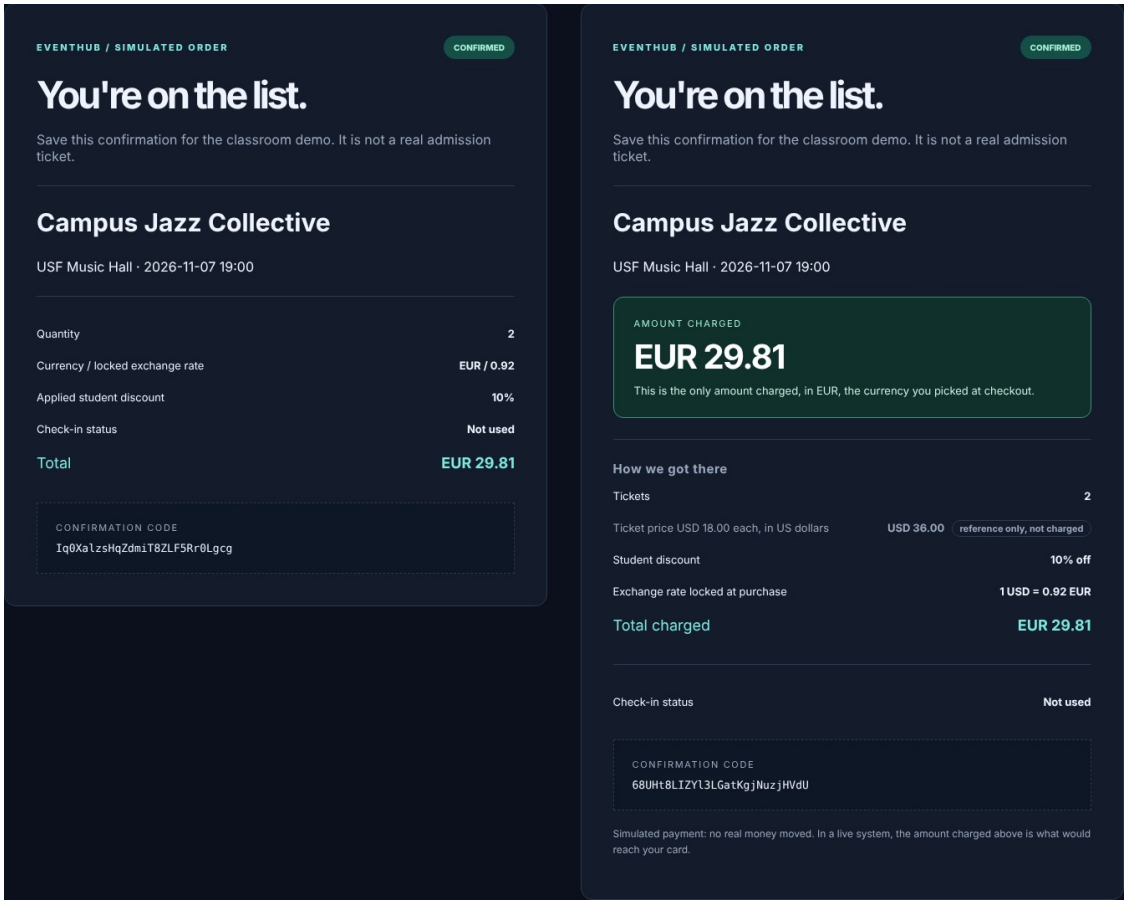



*Figure 1. The same purchase (2 tickets, EUR, student account) on the Part 4 build (left) and now (right).*

## 1.2 Requirement changes

All earlier IDs are kept. Changed requirements are superseded with reciprocal links so the course script still accepts them (107 requirements, 14 links, 0 failures).

| New or changed | What it says, in short | Why |
|---|---|---|
| purchase-024..028 | Every form carries a one-time token. A repeat returns the original order. Same token for a different purchase is refused. No token is refused. The button disables itself on submit. | Thuy: double buy undefined |
| purchase-013 → 029<br>purchase-017 → 030 | Confirmation shows the amount charged and its currency as the biggest figure. Every other amount is labelled "reference only". History shows the same wording. | Darnell: which amount hit my card |
| purchase-031, identity-018/019, pricing-016/017 | Currency chosen at registration (USD preselected), shown and changeable in the header on every page, explained on checkout, named next to the quick-buy hint. | Darnell: where does the default come from |
| purchase-032 [TBD Q19], audit-007 | Over the request limit: refuse with a wait time, create nothing, write one audit row per blocked period. | Kenji: request speed |
| audit-003 → 008 | "Exactly one audit record per attempt" now says attempts that reach the purchase path, including a repeated token. | Rate-limited requests never reach it |
| pricing-015, identity-020 [TBD Q18] | Eligibility is judged from the current email at every pricing. The website offers no email change in version one. | Kenji: discount and email |
| identity-017 | A registration whose email is not one address with one "@" and a dotted domain is refused. | Found by testing: `a@b@usf.edu` registered |

| New or changed | What it says, in short | Why |
|---|---|---|
| system-009 | Every automated test carries a requirement ID. | Thuy: tests trace to the spec |

## 1.3 Model and code changes

The C4 model gains one component, **Request Rate Limiter**, between Boundary Validation and Purchase & Orders. Purchase & Orders gains idempotent behaviour inside its existing transaction, with no new component. `workspace.dsl` is updated, the Call 4 version is kept in `docs/part4-design-baseline/`, and `Part5Components` and `Part5PurchasePath` are new Graphviz renderings (not Structurizr Lite exports, as in Part 4).

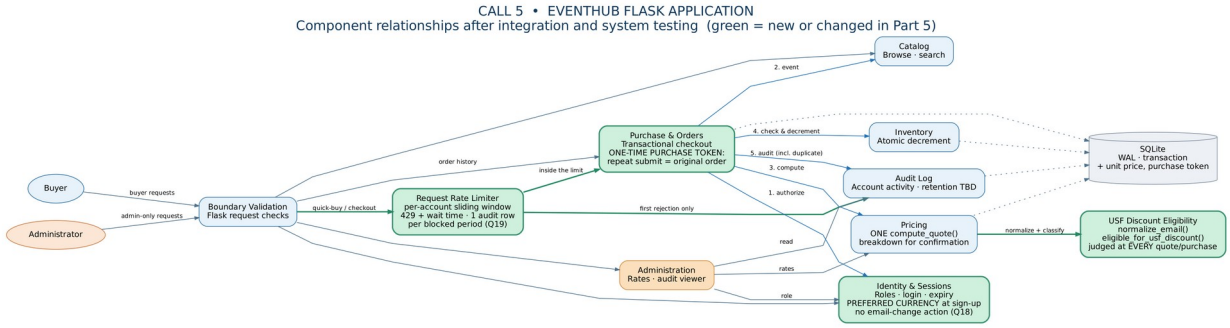



*Figure 2. Component view after Call 5. Green is new or changed.*

| File | Change |
|---|---|
| `eventhub/purchase.py` | `place_order()` checks the purchase token inside the write lock. Duplicate returns the original order and records `duplicate_suppressed`. `purchase()` stays as a wrapper so old callers work. |
| `eventhub/ratelimit.py` (new) | Per-account sliding window. Rejected requests are not counted, so hammering does not extend the block. Only the first rejection of a period is flagged for the audit log. |
| `eventhub/web.py`, templates, `static/app.js` | Token in every purchase form; 429 with `Retry-After`; header currency form; registration currency; one eligibility rule on the checkout notice; single-submit guard. |
| `eventhub/db.py` | Adds `request_token` and `unit_usd_minor` to orders and a unique index. `migrate()` upgrades a Part 4 database; a test proves an old order still opens. |
| `eventhub/pricing.py` | `order_breakdown()` gives the confirmation its facts. It adds no price arithmetic, so there is still one place that computes money. |
| `eventhub/identity.py` | Currency at registration; email shape check; PBKDF2 pin restored. |

**Decisions the team made without the customer** are marked as provisional everywhere: 20 requests per minute per account (Q19), "a repeat shows the original order" (Q20), "choose currency at registration" (Q21). **Decisions we did not make:** the performance number (Q13), whether emails can change (Q18), how to verify a USF address (Q3). `docs/risk-register.md` lists five risks with proposed decisions and the person who would sign. All are marked OPEN. Kenji said his findings block release, so until someone signs, they do.

## 1.4 Call 5 exit criteria, honestly

| Thuy's exit criterion | Status |
|---|---|
| Every requirement-linked test passes | Met. 281 of 281; 97 of 100 active requirements have a test, 3 have a written reason they cannot |
| Kenji's two findings fixed or formally accepted | Not met. Finding 2 is fixed with placeholder numbers. Finding 1 does not reproduce, but the real hole (RISK-001) needs the customer's signature |
| Confirmation page reworded | Met in code and tests. Not re-tested with real students (RISK-005) |
| Double-purchase behaviour defined and tested | Met, awaiting customer confirmation (Q20) |
| The performance sentence exists | Not met. Marcus owes it. We measured and report numbers without a verdict |

# 2. How the testing process went (60%)

## 2.1 What the GenAI did, and what I gave it

The GenAI tool was Claude (Sonnet 5.5) in a Claude Code session. My one instruction was to do Part 5 in the same way as the earlier parts, starting from the Part 4 package. Claude then did the whole job: read the code and requirements, ran the baseline, wrote the changes, wrote the tests, ran them and wrote this report. So the assignment's picture of a student prompting a GenAI and passing the result to Selenium happened here inside one session. **You should read the tests and the report yourself before submitting;** nobody else has.

The sequence, which matters because order changed what was found:

- **Baseline first.** Part 4 as delivered: 59 passed; Ruff format failed on 20 files; no PBKDF2 pin; no code that changes an email.
- **Probe the Call 5 scenarios on the old build** (two scripts, saved in `report/baseline/`). This overturned two of my own assumptions (see 2.5).
- **Black-box list from the requirements and transcript only**, using equivalence classes, boundaries and error guessing (`docs/test-plan.md`). Claude had already read the source, so the separation is one of method, not of knowledge.
- **White-box list from the source:** every branch, setting, transaction and template that could go wrong (same file).
- **Run once as generated, save the output untouched,** then diagnose.

## 2.2 Did the generated tests run without errors? No.

**Before any test ran,** the environment failed. The sandbox has Chrome 153 but ChromeDriver 147. Selenium Manager could not download a matching driver (the network allows package registries only), and a PyPI driver package tried to download the binary at build time and failed. What worked was `chromedriver-py`, which ships the driver inside the wheel. The fixture now uses `CHROMEDRIVER`, then `chromedriver-py`, then Selenium Manager, so a normal machine needs nothing special.

**First full run, nothing edited:** 137 tests collected, **110 passed and 27 failed** in 2 min 40 s. The raw output is `report/first_run/first_run_output.txt` and the tests exactly as generated are in `report/first_run/tests_system_as_generated/`. **All 27 failures were defects in my tests. None was a bug in the application.**

| Class | N | Root cause | Fix | Test or code? |
|---|---|---|---|---|
| A. Click intercepted | 13 | Buttons sat below the visible area, and `html{scroll-behavior:smooth}` animates every scroll, so Selenium clicked while the page was still moving | `click()` helper scrolls instantly to mid-screen | test |
| B. Option not found | 10 | Admin page options are `<option>GBP</option>` with no `value=`; `select_by_value` needs the attribute | select by visible text | test |
| C. Link not found | 1 | Link text is `Get tickets <span>↗</span>` | partial link text | test |
| D. Duplicate account | 1 | `grad@usf.edu` and `GRAD@USF.EDU` are one address in a shared database | unique prefix, same case and domain | test |
| E. Stale element | 1 | Reading page text during a navigation | waits retry on stale/missing element | test |
| F. Triple click | 1 | `ActionChains` addressed the card after navigation began | dispatch three real click events | test |

The path to green was not straight. My first fix for class A (a plain `scrollIntoView`) did nothing, and 17 tests still failed. Looking at the numbers, the click points were still far off screen, which pointed at the smooth scrolling in `style.css`. Instant scrolling fixed the helper, but the rest of the tests still called `.click()` directly, so 10 more failed the same way (class G). A sign-out test that had passed in the first run then failed (class H): it navigated away before the sign-out finished, and had passed only by timing luck.

## Evidence A. The root cause of the biggest cluster

```
# before (my first attempt)
driver.execute_script("arguments[0].scrollIntoView({block: 'center'});", element)

# after: style.css sets scroll-behavior: smooth, so ask for no animation
driver.execute_script("arguments[0].scrollIntoView({block: 'center', behavior:
'instant'});", element)
```

| Run | Result |
|---|---|
| First full run, as generated | 110 passed, 27 failed |
| After fixes A to F (targeted re-run of 32 tests) | 17 failed. Root cause of A was wrong |
| After the smooth-scroll fix | 11 failed. Direct `.click()` calls (G) and the sign-out race (H) |
| After routing every click through the helper | 24 of 24 selected tests passed |
| Second full run | 137 passed, 0 failed (171 s) |
| Final run after formatting and the one-button runner | 137 passed, 0 failed (174 s), plus 144 unit and integration |

## 2.3 Defects I caught in my own drafts before the first run

Twice I wrote a vacuous assertion (`assert ... or True`), the exact thing Thuy warned about: a test that cannot fail. I noticed both while re-reading and removed them before the first run, so they

are not in the 27. The traceability check also found nine requirements with no test at all, which I then wrote tests for rather than exempt.

## 2.4 What the tests found about the application

Being honest about this: **the browser suite found no new application bug on its first run.** What it did find came earlier, from reading and from probing:

- Replaying one form creates two orders (proved on Part 4). Fixed.
- A malformed email like a@b@`usf.edu` registers. Fixed (identity-017).
- The checkout "discount pending" notice used its own `endswith("@usf.edu")` instead of the one eligibility function, so the two could disagree. Fixed and pinned by a test.
- Registration had no currency choice and `/currency` was unreachable. Fixed.
- Part 4's code did not match its report (format, PBKDF2). Fixed.

## 2.5 Two of my assumptions were wrong

**"A double-clicked button creates two orders."** I wrote this into my first traceability table before testing it. On the Part 4 build it is false for the double-click and triple-click gestures: only one request reached the server. I corrected the table. The consequence is worth stating: **my browser double-click tests pass on the old code too.** They guard against regression but do not prove the bug. The tests that fail on Part 4 are the ones that send the same form twice from outside the page (`test_even_with_the_page_script_bypassed...`) and the unit tests for the token. I did not run the new suite against Part 4; I ran two probe scripts.

**"Kenji's discount bug is real."** I looked for the email-change code to fix and there is none. The finding does not describe our build. I did not invent an email-change feature to fix it; I wrote the property down, tested it and logged the real hole.

## 2.6 Do the tests actually fail when the code breaks?

Thuy's bar is a test that would fail if the behaviour broke. `scripts/mutation_check.py` breaks the code 20 ways, one at a time, in a throw-away copy (for example: remove the duplicate check, ignore the rate limit, round yen down, compare email domains with `endswith`, cache eligibility per account, drop the CSRF check). **All 20 are caught, each by a test that names the requirement it guards.**

The first version of this check was flattering and wrong. Two mutants crashed the whole app (a bad SQL placeholder, an unimported constant), which proves nothing. After `ruff format` rewrote the source, eight mutants silently stopped applying and the script reported 12 of 12. I rewrote the mutants to be realistic, made an unapplied mutant a hard failure, and required a test that names the requirement to be among those that fail. That exposed a real gap: the lookalike-domain rule (`mail.usf.edu`, `notusf.edu`) was tested only on the function, never through the purchase path, and the fast suite caught that mutant only by accident. The Selenium tests catch it properly (I checked by running them against the mutant). I then added a fast test for it.

## 2.7 Load measurement

20 signed-in buyers, 5 purchases each, at the same moment (`tests_system/test_performance.py`, plain HTTP so it measures the server, not Chrome). 100 purchases, 0 failures, nothing oversold. Purchase request latency: **median 22 to 41 ms; 95th**

**percentile 381 to 496 ms over six repeat runs; slowest 570 to 776 ms.** With 50 buyers the 95th percentile was about 1,150 ms.

**This test passes or fails against 500 ms, which is a number the team made up** (Q13). One of six runs missed it by 4 ms. Please treat the load test as a measurement. If it fails on your machine, change `EVENTHUB_PERF_P95_MS` or wait for Marcus's sentence. It is a development server on one sandbox, so it says nothing about production.

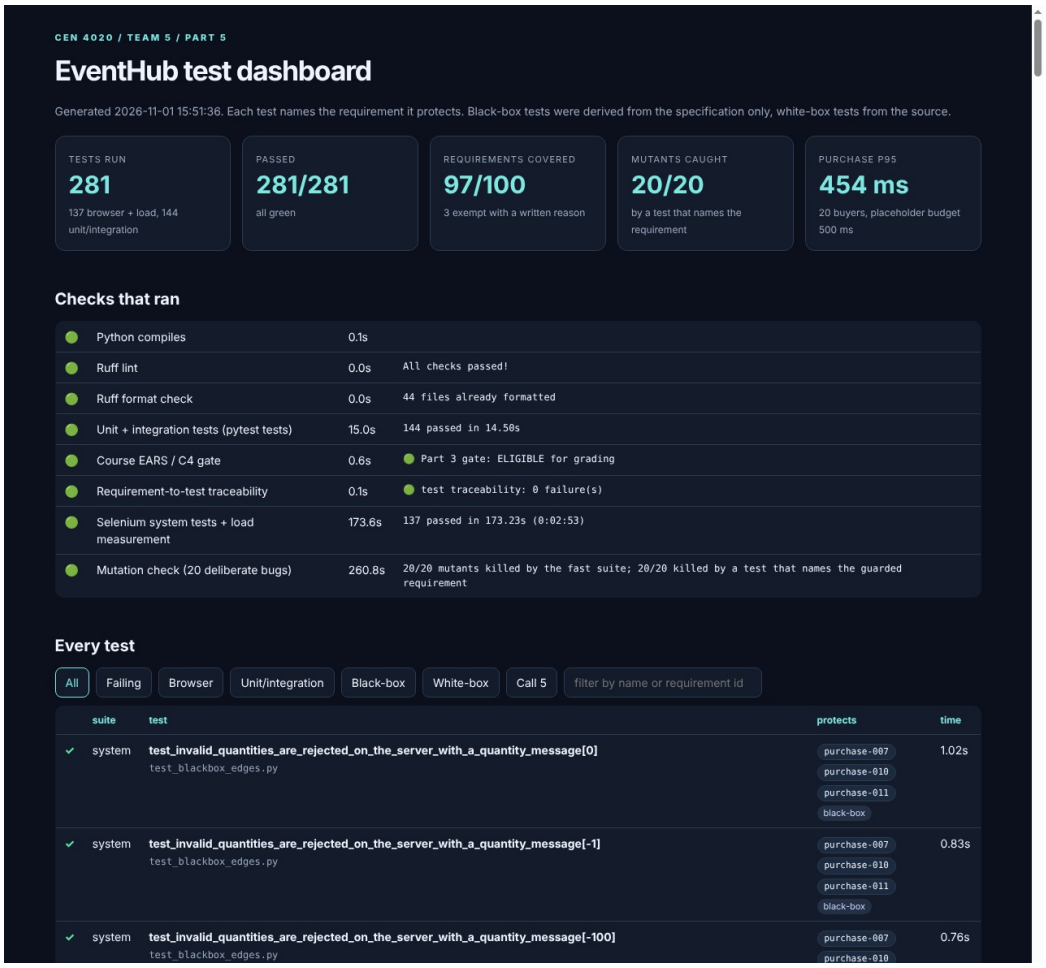



*Figure 3. `report/test_dashboard.html`, produced by `run_all_tests.py`. Every test is filterable and shows its requirement IDs.*

## 2.8 What I would do differently

- **Run one test through the browser before writing 137.** Classes A to C would have shown up in minutes and been fixed once, not 24 times.
- **Probe the old build before writing the traceability table,** not after.
- **Ask the GenAI what would make each test fail,** not just what to test. The mutation check was the most useful single idea in the part.
- **Give the GenAI the markup,** not only the requirements. Three of six failure classes were wrong guesses about the page (`<option>` without `value`, a nested `<span>`, smooth scrolling).
- **Keep the customer's numbers out of the tests.** Where a number is unknown, the test reports instead of judging.

## 3. Tests run against the final application (15%)

The system tests start the real application with `eventhub.web.create_app()` (the same call as `run.py`, which a unit test confirms) in a subprocess with a fresh database, and drive it with Selenium. Three extra servers cover settings the default one cannot (no discount configured, a purchase cap, a 3-second session timeout). To run everything:

```
python -m pip install -r requirements-dev.txt -r requirements-system.txt
python run_all_tests.py                    # about 4 minutes, needs Chrome
python run_all_tests.py --with-mutation    # also breaks the code on purpose (about 8
minutes)
```

| Step | Final result |
|---|---|
| Compile; Ruff lint; Ruff format check | passed (44 files formatted) |
| Unit and integration tests | 144 passed in 14.5 s (Part 4 had 59) |
| Course EARS / C4 gate | ELIGIBLE for grading (107 requirements) |
| Requirement-to-test traceability | 0 failures; 97 of 100 active requirements covered, 3 exempt with reasons |
| Selenium system tests and load measurement | 137 passed in 173 s |
| Mutation check | 20 of 20 caught by a test that names the requirement |

**Limits:** one Linux sandbox, headless Chrome 153 at desktop size plus a 375 px layout check. Not macOS or Windows (the Part 4 Mac failure was exactly that kind of difference). The rate limiter is per process. The workflow file is untested on GitHub. Back-button restores from the browser cache are not automated.

## 4. The test cases (5%)

Every test carries `@pytest.mark.req(...)`; the suite refuses to collect one that does not. `docs/system-test-catalog.md` lists all 281 with their requirements and results. By file:

| File | N | What it covers |
|---|---|---|
| `test_e2e_journey.py` | 14 | Thuy's full journey (register, sign in, search, open, buy in EUR, confirmation, history); sign-in return; sign-out; double-click quick-buy; abandoned checkout (cancel and walk away); Back button; phone layout |
| `test_blackbox_edges.py` | 73 | 12 bad quantities, stock boundaries, 6 bad currencies, 4 currencies bought, logged-out access, search edge cases and XSS payloads, registration limits, login, open redirects, tampered CSRF, roles, other buyer's order, unknown pages |
| `test_whitebox_edges.py` | 29 | Yen rounding, estimate equals charge, USF address variants, no-discount server, rate change vs old orders, bad rates, audit rows, check-in once, cap, session timeout, headers, oversized body |
| `test_call5_findings.py` | 19 | Double buy (UI and replay), token rules, 40-request hammer, confirmation wording and font size, currency visibility, no email change, discount follows email, RISK-001 characterization |
| `test_oversell_browsers.py` | 1 | Two real browsers press buy together on the last ticket, 5 rounds: exactly one wins, one is told "sold out" |
| `test_performance.py` | 1 | Load measurement (Section 2.7) |

| File | N | What it covers |
|---|---|---|
| `tests/` (16 files) | 144 | Components, HTTP, idempotency, rate limiter, eligibility timing, 200-round oversell, 30 buyers for 5 tickets, Part 4 database upgrade, 9 previously untested requirements |

Of the 137 browser and load cases, 73 are tagged black-box, 29 white-box, and 60 Call 5 across both suites.

## Before we submit

- Read the tests and this report. Change the author line if the team wants it different.
- Get Kenji, Darnell and Marcus to answer RISK-001 to RISK-005 and Q18 to Q21. Do not claim acceptance until then.
- Run `python run_all_tests.py` on a Mac, and push to GitHub to see the workflow run.
- Repeat Darnell's student-union session with the new confirmation page.